

Case 14: Negative — GET with Invalid Query Parameters

Intermediate+

Verify the API handles invalid query parameter values correctly.

New concept: Invalid query parameter values

API Endpoint

GET `https://jsonplaceholder.typicode.com/posts?userId=abc`

What You Will Learn

- Passing invalid values as query parameters
- String where integer expected in query params
- Empty query parameter values
- Parametrizing multiple invalid query param values

Study Plan

Step 1 — What are invalid query params?

In Case 7 you sent `params={'userId': 1}` which is valid. What happens if you send `userId='abc'` (string), `userId=""` (empty), or `userId=-1` (negative)?

Step 2 — Build invalid param values

Create a list of invalid `userId` values: `'abc'` (string), `"` (empty string), `-1` (negative), `99999` (non-existent user). Each should cause unexpected behavior.

Step 3 — How to pass invalid params

Same as Case 7: `requests.get(url, params={'userId': invalid_value})`. The changing value is the `userId` value, not the URL.

Step 4 — What to assert

For invalid types like `'abc'`, a real API returns 400. For non-existent `userId` like `99999`, a real API returns 200 with an empty list `[]`. Think: which needs `xfail` and which doesn't?

Step 5 — Think about the loop

For `userId=99999`, the response is a valid list but empty. Do you need a for loop to verify it's empty, or is `len(data) == 0` enough?

Think Before You Code

Ask yourself these questions before writing a single line:

- What is the changing value between each test run — the URL or the query param value?
- For `userId=99999`, what type does `response.json()` return — a dict or a list?
- Do you need a for loop to check if the list is empty?
- Which invalid values need `xfail` and which can be asserted directly?
- How is `params={'userId': 'abc'}` different from `params={'userId': 99999}`?

Your Task

Write pytest tests for invalid query parameter values on GET /posts.

Test A — Non-existent userId (should return empty list):

params={'userId': 99999}

Assert: status 200, response is a list, list is empty

Test B — Parametrized invalid types (should return 400):

Test these userId values: 'abc', '', -1

Mark as xfail — JSONPlaceholder may not reject these

Assert: status 400

Before writing:

- What is the parametrize variable in Test B?
- For Test A, do you need a loop or is `len(data) == 0` enough?
- What type does `response.json()` return for an empty result?

BONUS: Print `response.json()` for each invalid type — what does JSONPlaceholder actually return for `userId='abc'`?

Hint: Two separate test functions — one for empty list (no parametrize), one for invalid types (parametrize). For empty list: `assert len(data) == 0`, no loop needed.